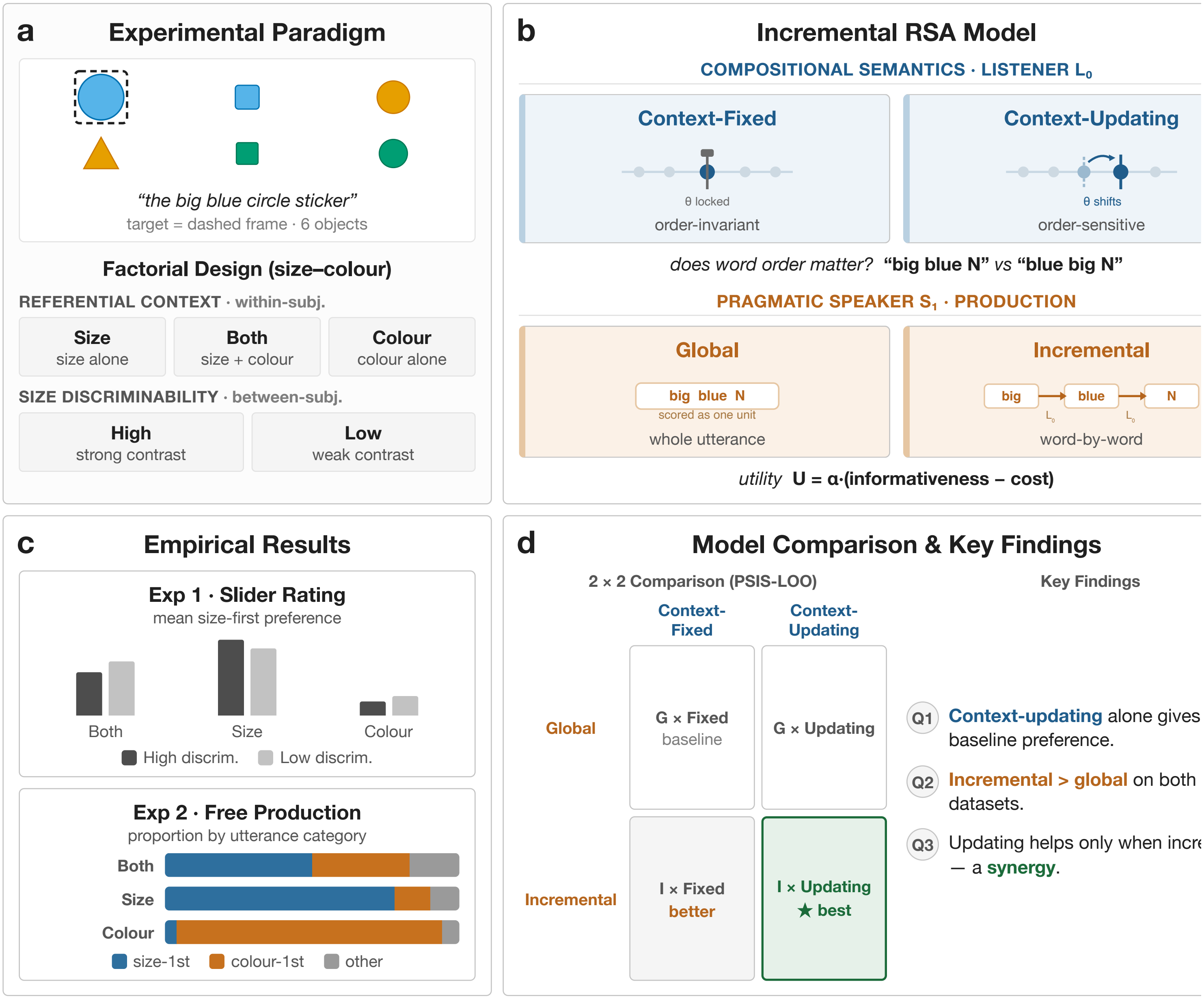
a
Experimental Paradigm
"the big blue circle sticker"
target = dashed frame · 6 objects
Factorial Design (size–colour)
REFERENTIAL CONTEXT · within-subj.
Size
size alone
Both
size + colour
Colour
colour alone
SIZE DISCRIMINABILITY · between-subj.
High
strong contrast
Low
weak contrast
b
Incremental RSA Model
COMPOSITIONAL SEMANTICS · LISTENER L_0
Context-Fixed
θ locked
order-invariant
Context-Updating
θ shifts
order-sensitive
does word order matter? "big blue N" vs "blue big N"
PRAGMATIC SPEAKER S_1 · PRODUCTION
Global
big blue N
scored as one unit
whole utterance
Incremental
big
blue
N
L_0
L_0
word-by-word
utility U = α·(informativeness – cost)
c
Empirical Results
Exp 1 · Slider Rating
mean size-first preference
Both
Size
Colour
High discrim.
Low discrim.
Exp 2 · Free Production
proportion by utterance category
Both
Size
Colour
size-1st
colour-1st
other
d
Model Comparison & Key Findings
2 × 2 Comparison (PSIS-LOO)
Context-Fixed
Context-Updating
Global
Incremental
G × Fixed
baseline
G × Updating
I × Fixed
better
I × Updating
★ best
Key Findings
Q1 Context-updating alone gives baseline preference.
Q2 Incremental > global on both datasets.
Q3 Updating helps only when incre
— a synergy.